

# SwapBuds Complete Launch Plan

## Master TODO Lists & Pre-Launch Checklists

### PART 1: EXECUTIVE SUMMARY

#### ▮ Project Overview

- **Application:** SwapBuds - P2P Trading Marketplace for Sustainable Item Swapping
- **Tagline:** "Swap More, Buy Less"
- **Launch Date:** March 17, 2026
- **Timeline:** 16 weeks (November 22, 2025 - March 16, 2026)
- **Solo Developer:** Darius Bere (Standout Development SRL)
- **Work Schedule:** 20-25 hours/week, evenings/weekends

#### ▮ Key Metrics by Milestone

| Milestone        | Week | Target           | Key Deliverables                   |
|------------------|------|------------------|------------------------------------|
| Legal Foundation | 4    | Legal docs ready | Registration with age verification |
| MVP Complete     | 8    | Free tier live   | All features functional, deployed  |
| Beta Launch      | 12   | 10-20 beta users | First real trades completed        |
| Public Launch    | 16   | 500 users        | Platform live, marketing push      |

#### ▮ Target Audience

Collectors, gamers, anime fans, sustainability enthusiasts, Gen Z/Millennials, eco-conscious communities

### PART 2: TECHNICAL STACK & DEPLOYMENT

#### Current Status

- ✓ Backend v1.0: Complete with 484 passing tests
- ⚠ Frontend v0.2: Auth UI foundation (90% remaining work)
- ✓ Planning: Financial, Legal, Marketing, Deployment docs complete

#### Technology Stack

##### Backend

- Runtime: Node.js v18
- Framework: NestJS
- Database: PostgreSQL (Prisma ORM)
- Cache: Redis
- Real-time: WebSockets
- File Upload: Cloudinary
- Email: Resend or AWS SES

- Monitoring: Sentry

## Frontend

- Framework: Next.js (React)
- Language: TypeScript
- Styling: Tailwind CSS
- Package Manager: Yarn
- Deployment: Vercel or Cloudflare Pages

## Infrastructure Timeline

### Phase 1: Testing (Weeks 1-12) - FREE

| Service          | Cost               | Limits                   | Use Case                     |
|------------------|--------------------|--------------------------|------------------------------|
| Vercel Backend   | Free               | Serverless, cold starts  | Testing environment          |
| Neon Database    | Free               | 512MB, 5 hrs/day compute | Development & testing        |
| Upstash Redis    | Free               | 10K requests/day         | Cache, rate limiting         |
| Resend Email     | Free               | 3K/month, 100/day        | Testing notifications        |
| Cloudinary       | Free               | 25GB storage/bandwidth   | Image uploads                |
| Cloudflare Pages | Free               | Unlimited bandwidth      | Frontend hosting             |
| Sentry           | \$26/month         | 5K errors/month          | Error monitoring (essential) |
| <b>Total</b>     | <b>~\$26/month</b> | Limited but sufficient   | MVP validation               |

### Phase 2: Production (Week 13+) - \$150-220/month

| Service      | Cost                   | Limits                 | Recommendation                  |
|--------------|------------------------|------------------------|---------------------------------|
| Backend      | \$7-19                 | 512MB-2GB RAM          | Railway or Render               |
| Database     | \$19                   | 3GB, unlimited compute | Neon Scale                      |
| Redis        | \$10-20                | Autoscaling            | Upstash or Railway              |
| Email        | \$10-30                | 10K+ sends/month       | AWS SES or Resend               |
| Storage      | \$89                   | 500GB+ bandwidth       | Cloudinary Plus                 |
| Monitoring   | \$26-52                | Unlimited errors       | Sentry Team                     |
| Uptime       | \$10-18                | Multi-location         | BetterStack                     |
| Frontend     | \$0-20                 | Unlimited              | Cloudflare (free) or Vercel Pro |
| <b>Total</b> | <b>\$171-268/month</b> | Production-grade       | Scalable & reliable             |

## PART 3: AUTOMATION TOOLS & PROCESSES (BEFORE GO-LIVE)

### ✓ Recommended Automation Setup

#### CI/CD & Code Quality (FREE)

##### 1. GitHub Actions (Already in [DEPLOYMENT.md](#))

- Backend CI workflow: `backend-ci.yml`
  - Run 484 tests on every PR
  - Lint check, TypeScript validation
  - Generate Prisma client
  - Code coverage reporting
- Frontend CI workflow: `frontend-ci.yml`
  - Run linting and type check
  - Build production bundle
  - Run E2E tests
  - Lighthouse CI performance monitoring
- Auto-deploy to production on merge to main
- **Setup time:** 2 hours
- **Value:** Catch bugs before deployment

##### 2. Dependabot (FREE)

- Automatic dependency update PRs
- Security vulnerability alerts
- **Setup time:** 30 minutes
- **Benefit:** Stay on latest security patches

##### 3. GitHub Secret Scanning (FREE)

- Detect exposed secrets automatically
- **Setup time:** 15 minutes
- **Benefit:** Prevent accidental secret commits

#### Monitoring & Error Tracking

##### 4. Sentry (\$26/month Team plan)

- Backend error tracking
- Frontend error tracking
- Source map uploads for debugging
- Session replays for debugging issues
- **Setup time:** 2 hours
- **Critical for production stability**

##### 5. BetterStack (\$10/month - Uptime monitoring)

- Website uptime monitoring from 10 locations
- Incident management dashboard
- SMS/Email/Slack alerts when down
- Status page for users
- **Setup time:** 1 hour
- **Peace of mind for solo developer**

##### 6. Grafana Cloud (FREE tier)

- Metrics collection and visualization
- Log aggregation
- Alerting on anomalies
- **Setup time:** 2 hours
- **Better visibility into system health**

## Security & Performance

### 7. npm/Yarn Audit (Built-in)

- Run `yarn audit` to find vulnerabilities
- Integrate into CI/CD (required before deploy)
- **Frequency:** Every commit

### 8. Snyk (FREE tier)

- Continuous vulnerability scanning
- Automatic PR generation for fixes
- SAST (static analysis) for code issues
- **Setup time:** 1 hour
- **Proactive security**

### 9. Lighthouse CI (Already in frontend workflow)

- Performance & accessibility scoring
- Prevents regression
- **Frequency:** Every PR

## Database & Backup Automation

### 10. Neon Backups (Built-in)

- Automated daily backups
- 30-day retention
- Point-in-time recovery enabled
- **Verify:** Test restoration monthly

### 11. Cloudinary Backup (Built-in)

- Automatic backups in separate region
- Enable auto-versioning
- **Frequency:** Continuous

### 12. Environment Variables Management

- Use GitHub Secrets for all environments
- Separate secrets for dev/staging/prod
- Rotate critical secrets every 90 days
- **Tool:** GitHub Secrets Manager (built-in)

## ☐ Automation Checklist (Before Go-Live)

- ☐ GitHub Actions workflows created and tested
  - ☐ Backend CI/CD pipeline working
  - ☐ Frontend CI/CD pipeline working
  - ☐ Auto-deploy to production on main push
  - ☐ Rollback mechanism tested
- ☐ Dependabot configured

- ☐ Enabled for both repos
- ☐ Set to auto-merge patch updates
- ☐ Monitoring setup completed
  - ☐ Sentry connected to backend and frontend
  - ☐ BetterStack monitoring configured
  - ☐ Grafana dashboards created
  - ☐ Alerts configured for key metrics
  - ☐ Test alert channels work (email, Slack)
- ☐ Database backups tested
  - ☐ Neon backups verified
  - ☐ Restoration test completed successfully
  - ☐ Retention policy configured (30 days minimum)
- ☐ Security automation
  - ☐ GitHub secret scanning enabled
  - ☐ Snyk integration added
  - ☐ npm audit passing
  - ☐ SAST rules configured
- ☐ Logging & observability
  - ☐ Application logs flowing to Sentry
  - ☐ Database query logs accessible
  - ☐ Error rate alerts configured
  - ☐ Performance metrics tracked

**Total setup time: 12-15 hours | Critical for solo developer to maintain visibility & reliability**

## **PART 4: COMPREHENSIVE TODO LISTS BY ASPECT**

### **▯ TECHNICAL TODO LIST**

#### **Frontend Development (Weeks 2-6: ~160 hours remaining)**

##### **Week 2: Item Management UI (24 hours)**

- ☐ Create items listing page (grid view)
- ☐ Build ItemCard component (image, title, condition)
- ☐ Implement infinite scroll/pagination
- ☐ Create item detail page with image gallery
- ☐ Build item creation form with Cloudinary upload
- ☐ Add category & condition filters
- ☐ Implement search with debouncing
- ☐ Add estimated value field (optional)
- ☐ Create item edit form
- ☐ Add item deletion with confirmation
- ☐ Test responsive design on mobile

##### **Week 3: Item Management Continued + User Profiles (24 hours)**

- ☐ Create user profile page with stats
- ☐ Build profile edit form (avatar, bio, location)

- ☐ Implement avatar upload with cropping
- ☐ Create "My Items" tab on profile
- ☐ Build settings page structure
- ☐ Implement privacy settings section
- ☐ Add data export request button
- ☐ Add account deletion button with warnings
- ☐ Create cookie preferences modal
- ☐ Test profile/settings flows end-to-end

#### **Week 4: Trading System UI (24 hours)**

- ☐ Create trade proposal modal
- ☐ Implement full trade creation flow
- ☐ Build trades list page (incoming, outgoing, completed)
- ☐ Create trade detail page with timeline
- ☐ Add trade acceptance/rejection buttons
- ☐ Implement trade completion flow
- ☐ Add trade status updates display
- ☐ Add delivery method selection
- ☐ Create counter-offer UI (if multi-item trades)
- ☐ Add trade cancellation with confirmation
- ☐ Test all trade scenarios end-to-end

#### **Week 5: Messaging System UI (24 hours)**

- ☐ Create conversations list page
- ☐ Build conversation detail with chat UI
- ☐ Implement real-time messaging (WebSocket or polling)
- ☐ Add message input with send button
- ☐ Show typing indicators (if WebSocket)
- ☐ Add message notifications
- ☐ Display unread message count
- ☐ Link messages to trades/items
- ☐ Add conversation search/filter
- ☐ Optimize for mobile performance

#### **Week 6: Reviews, Notifications & Legal UI (20 hours)**

- ☐ Create review submission form (post-trade)
- ☐ Display user reviews on profile
- ☐ Show average rating & review count
- ☐ Create notifications page
- ☐ Implement notification bell with count
- ☐ Add notification types (trade, message, review, etc.)
- ☐ Mark notifications as read
- ☐ Add notification preferences
- ☐ Create age verification form
- ☐ Build ToS/Privacy acceptance UI
- ☐ Test notification flows

## **Testing & Quality Assurance (Weeks 7-9: ~72 hours)**

### **Week 7-8: Bug Fixes & Polish (24 hours)**

- ☐ Manual testing all features end-to-end
- ☐ Cross-browser testing (Chrome, Safari, Firefox)
- ☐ Mobile responsiveness verification
- ☐ Fix critical bugs
- ☐ Resolve UX/UI issues
- ☐ Romanian language check throughout
- ☐ Performance optimization
- ☐ Accessibility check (basic WCAG)
- ☐ SEO basics (meta tags, titles)
- ☐ Create error pages (404, 500)
- ☐ Test email notifications end-to-end
- ☐ Create user guide / FAQ

### **Week 9: Security & GDPR Final Check (24 hours)**

- ☐ Security audit (basic)
- ☐ Test auth flows for vulnerabilities
- ☐ Verify rate limiting on all endpoints
- ☐ Check file upload security
- ☐ SQL injection vulnerability scan
- ☐ Test CORS configuration
- ☐ Verify API authentication
- ☐ Check password hashing
- ☐ Test session management
- ☐ Test data export functionality
- ☐ Test account deletion functionality
- ☐ Verify age verification works
- ☐ GDPR compliance checklist complete

### **Week 9-10: Performance & Optimization (24 hours)**

- ☐ Database query optimization
- ☐ Redis caching implementation
- ☐ Database indexes verification
- ☐ N+1 query elimination
- ☐ Load testing simulation
- ☐ Performance metrics acceptable (<3s load, <200ms API)
- ☐ Image optimization
- ☐ Bundle size optimization
- ☐ Code splitting implementation
- ☐ Service worker deployment

## Deployment & Infrastructure (Weeks 8-10: ~36 hours)

### Week 8: Testing Environment Deployment (12 hours)

- ☐ Set up Railway or Render account
- ☐ Deploy backend to free tier
- ☐ Set up Neon PostgreSQL database
- ☐ Set up Upstash Redis cache
- ☐ Configure environment variables
- ☐ Deploy frontend to Cloudflare Pages
- ☐ Set up custom domain ([swapbuds.ro](https://swapbuds.ro))
- ☐ Configure DNS settings
- ☐ Set up SSL certificates
- ☐ Run database migrations on prod
- ☐ Seed initial data (categories, etc.)
- ☐ Test API endpoints in production

### Week 9: Monitoring & Alerting Setup (12 hours)

- ☐ Configure Sentry for backend errors
- ☐ Configure Sentry for frontend errors
- ☐ Set up BetterStack uptime monitoring
- ☐ Create Grafana dashboards
- ☐ Configure alert thresholds
- ☐ Test alert channels
- ☐ Set up error rate alerts (5%)
- ☐ Set up downtime alerts
- ☐ Set up slow query alerts
- ☐ Configure notification channels (email, Slack)

### Week 10: Backup & Disaster Recovery (12 hours)

- ☐ Configure automated database backups
- ☐ Set backup retention (30 days minimum)
- ☐ Test backup restoration process
- ☐ Set up backup monitoring/alerts
- ☐ Document restore procedure
- ☐ Create disaster recovery plan
- ☐ Define RTO (Recovery Time Objective: 4 hours)
- ☐ Define RPO (Recovery Point Objective: 24 hours)
- ☐ Document rollback procedures
- ☐ Test rollback process

## 📌 MARKETING TODO LIST



## **Pre-Launch Foundation (Weeks 1-4: ~60 hours)**

### **Week 1-2: Branding & Positioning**

- ☐ Finalize logo (color, white, black versions)
- ☐ Create brand guidelines document
- ☐ Design social media templates
- ☐ Create email templates
- ☐ Design banner images for all platforms
- ☐ Create app icons/favicons
- ☐ Design business cards (if needed)
- ☐ Develop brand voice/tone guide
- ☐ Finalize tagline options to test

### **Week 1-2: Website & Landing Page**

- ☐ Create marketing-focused homepage
- ☐ Write clear value proposition
- ☐ Create "How It Works" page (3-4 steps)
- ☐ Create "About Us" page
- ☐ Set up blog structure
- ☐ Implement Google Analytics
- ☐ Add social proof section
- ☐ Create email waitlist signup
- ☐ Design email confirmation page
- ☐ Add FAQ section

### **Week 1-2: Social Media Setup**

- ☐ Create Instagram account (@SwapBuds)
- ☐ Create TikTok account (@SwapBudsOfficial)
- ☐ Create Twitter/X account (@SwapBuds)
- ☐ Create Facebook page
- ☐ Create LinkedIn company page
- ☐ Create Discord server
- ☐ Set up YouTube channel
- ☐ Brand all profiles consistently
- ☐ Write compelling bios

### **Week 3-4: Content Creation & Sequences**

- ☐ Write launch press release
- ☐ Write 5 initial blog posts
- ☐ Create email welcome sequence (5 emails)
- ☐ Create social media caption bank (50 posts)
- ☐ Take product screenshots (10 angles)
- ☐ Create platform demo video (2-3 min)
- ☐ Create "How It Works" explainer (60 sec)
- ☐ Film tutorial videos (5 topics)
- ☐ Create social media graphics (20 templates)
- ☐ Design 3-5 infographics

### **Week 3-4: Tools & Platforms**

- ☐ Set up email marketing (Mailchimp or Resend)
- ☐ Set up social media scheduling (Buffer or Later)
- ☐ Set up design tools (Canva Pro: \$13/month)
- ☐ Configure analytics dashboard
- ☐ Set up customer support system
- ☐ Create Google Business Profile

### **Beta Program & Community Building (Weeks 5-10: ~80 hours)**

#### **Week 5-6: Beta Launch Preparation**

- ☐ Create beta tester application form
- ☐ Write beta welcome email
- ☐ Create beta testing guidelines
- ☐ Design beta tester badges/swag
- ☐ Set up feedback collection system
- ☐ Create private beta Discord channel
- ☐ Prepare beta launch announcement

#### **Week 6-7: Influencer Outreach**

- ☐ Create list of 50 micro-influencers (10K-100K followers)
- ☐ Focus: sustainability, minimalism, collectors, anime
- ☐ Draft influencer outreach email template
- ☐ Create influencer partnership package
- ☐ Reach out to first 10 influencers
- ☐ Follow up with interested parties
- ☐ Provide early access/beta features
- ☐ Schedule content calendar with partners

#### **Week 7-10: Community Engagement**

- ☐ Launch Discord server with channels
- ☐ Create welcome bot and rules
- ☐ Recruit 5 beta community moderators
- ☐ Host first community event/AMA
- ☐ Create community guidelines
- ☐ Set up community rewards program
- ☐ Publish first 3 blog posts
- ☐ Begin SEO optimization
- ☐ Submit to relevant directories
- ☐ Begin email newsletter (bi-weekly)

#### **Week 10-11: Partnership Outreach**

- ☐ Identify 10 potential community partners
- ☐ Contact local collecting clubs
- ☐ Reach out to university sustainability groups
- ☐ Explore convention partnerships
- ☐ Connect with online collecting forums

- ☐ Contact local thrift stores
- ☐ Reach out to eco-friendly brands

## **Launch Week & Scaling (Weeks 12-16: ~100 hours)**

### **Week 11-12: Advertising & Press Setup**

- ☐ Create Facebook/Instagram ad account
- ☐ Design ad creative (5 variations)
- ☐ Write ad copy (10 variations)
- ☐ Set up Google Ads campaign
- ☐ Configure conversion tracking
- ☐ Create retargeting pixels
- ☐ Plan ad budget allocation
- ☐ Create comprehensive press list (50 contacts)
- ☐ Write pitch emails (3 variations)
- ☐ Reach out to tech publications
- ☐ Contact collectibles/gaming media
- ☐ Pitch to 10 podcasts
- ☐ Schedule interviews for launch week
- ☐ Create media kit landing page

### **Week 12: Launch Preparation**

- ☐ Review website copy one final time
- ☐ Test all user flows
- ☐ Optimize page load speeds
- ☐ Mobile responsiveness check
- ☐ Set up customer support channels
- ☐ Create support documentation
- ☐ Publish press release on PRWeb
- ☐ Send launch email to waitlist
- ☐ Prepare social media posts (scheduled)
- ☐ Prepare Reddit AMAs for Day 2
- ☐ Line up Product Hunt supporters

### **Launch Week (Week 13): Execution**

- **Day 1 (Monday morning):**
  - ☐ Remove "Coming Soon" badges
  - ☐ Go live on Product Hunt
  - ☐ Post launch announcement all platforms
  - ☐ Send launch email to waitlist
  - ☐ Post in Reddit communities
  - ☐ Post in Facebook groups
  - ☐ Share on LinkedIn
- **Day 1 (Afternoon/Evening):**
  - ☐ Post first blog post
  - ☐ Launch ad campaigns (small budget)
  - ☐ Monitor social media responses

- ☐ Respond to all comments
- ☐ Check analytics
- **Days 2-5:**
  - ☐ Influencer content goes live
  - ☐ Share user-generated content
  - ☐ Post tutorials on YouTube/TikTok
  - ☐ Host Reddit AMAs
  - ☐ Respond to press inquiries
  - ☐ Monitor and fix technical issues
  - ☐ Host live stream QA
  - ☐ Discord community events
  - ☐ Feature Trader of the Day
- **Week 2-4:**
  - ☐ Continue social media engagement
  - ☐ Monitor analytics and adjust
  - ☐ Collect user feedback
  - ☐ Fix bugs reported by users
  - ☐ Implement top feature requests
  - ☐ Share success stories
  - ☐ Expand beta to 100+ users
  - ☐ Scale to 500+ users by week 4

## ▮ FINANCIAL TODO LIST

### Banking & Financial Setup

#### Before Launch (By Week 1):

- ☐ Separate business bank account created
- ☐ Business account at Romanian bank
- ☐ Business debit card ordered
- ☐ Business credit card for expenses (optional)
- ☐ Business accounting software set up (QuickBooks, Wave, or similar)
- ☐ Digital receipt storage system configured

### Accounting & Bookkeeping

#### Monthly (Start from Week 1):

- ☐ Record all income (premium subscriptions, features, etc.)
- ☐ Record all business expenses
- ☐ Categorize expenses properly
- ☐ Reconcile bank statements
- ☐ Monitor cash flow

#### Quarterly:

- ☐ Review financial statements
- ☐ Analyze metrics (CAC, LTV, etc.)

- ☐ Prepare quarterly tax payments if required
- ☐ Review budget vs. actual spending

**Annually:**

- ☐ Prepare annual income statement
- ☐ Prepare annual balance sheet
- ☐ Prepare annual cash flow statement
- ☐ File annual tax returns
- ☐ Calculate and pay annual taxes
- ☐ Review financial strategy for next year

**Taxation & Compliance**

**Year 1 (Before Public Launch):**

- ☐ ANAF registration (CUI code assignment)
- ☐ Understand TVA/VAT requirements
- ☐ Determine if TVA registration needed (probably yes for SRL)
- ☐ Set up tax estimation and payment schedule
- ☐ Consult with accountant on tax strategy
- ☐ Understand self-employed tax obligations

**Ongoing:**

- ☐ Track VAT from incoming money
- ☐ Track VAT from outgoing payments
- ☐ Calculate quarterly VAT payments if needed
- ☐ Prepare VAT return filings
- ☐ Monitor changing tax laws

**Pricing & Monetization**

**Before Year 2 Transition:**

- ☐ Finalize premium tier pricing (\$4.99-9.99/month vs. annual)
- ☐ Decide on Featured Listings pricing (\$2.99 for 7 days)
- ☐ Determine Organization Account pricing (\$49-99/month)
- ☐ Plan monetization timeline (Month 13 or later)
- ☐ Create pricing page copy
- ☐ Set up payment processing (Stripe recommended)
- ☐ Test payment flows end-to-end

**Financial Forecasting**

**Monthly:**

- ☐ Update revenue projections
- ☐ Track key metrics (CAC, LTV, retention)
- ☐ Analyze customer acquisition costs
- ☐ Monitor burn rate vs. projections
- ☐ Adjust budget allocations

**Quarterly:**

- ☐ Review 3-month performance
- ☐ Adjust annual projections
- ☐ Plan for next quarter spending
- ☐ Identify cost savings opportunities

## 🔑 LEGAL TODO LIST

### Business Entity & Registration

#### Before Launch (Critical):

- ☒ SRL formed at ONRC (Standout Development SRL - assuming done)
- ☐ ANAF registration completed
- ☐ CUI (Cod Unic de Înregistrare) assigned
- ☐ Tax ID received
- ☐ Business address registered
- ☐ Legal representative confirmed

### Accounting & Compliance

#### Before Launch:

- ☐ Contract with Romanian accountant (contabil autorizat)
- ☐ Set up monthly reporting to accountant
- ☐ Open separate business bank account
- ☐ Business insurance quotes obtained

#### Ongoing:

- ☐ Monthly accounting reports to accountant
- ☐ Annual financial statements preparation
- ☐ Tax compliance reviews quarterly
- ☐ Accounting records organized and backed up

### Legal Documents (All in Romanian)

#### Critical - Before Public Launch:

- ☐ Terms of Service (TOS)
  - ☐ Use iubenda template as base
  - ☐ Customize for SwapBuds specifics
  - ☐ Romanian language verified
  - ☐ Legal review by lawyer (€200-400)
  - ☐ Published on website
  - ☐ Users must accept at registration
- ☐ Privacy Policy
  - ☐ GDPR-compliant template
  - ☐ Romanian language translation
  - ☐ Clear data collection statement
  - ☐ Data retention policy defined
  - ☐ Third-party services disclosed

- ☐ Legal review by lawyer
- ☐ Published on website
- ☐ Linked in footer
- ☐ Cookie Policy
  - ☐ Generate with cookie management tool (iubenda, CookieYes)
  - ☐ List all cookies used
  - ☐ User consent implementation
  - ☐ Cookie banner displays on first visit
  - ☐ Romanian language verified
  - ☐ Published on website
- ☐ Community Guidelines / Code of Conduct
  - ☐ Define acceptable trading practices
  - ☐ Define prohibited items/behavior
  - ☐ User reporting procedures
  - ☐ Enforcement and moderation process
  - ☐ Appeal process
  - ☐ Published in app and website

## GDPR Compliance (Mandatory for EU)

### Before Launch - Implementation:

- ☐ Age verification (18+)
  - ☐ Date of birth field in registration
  - ☐ Age calculation logic
  - ☐ Reject if under 18
  - ☐ Testing: verify under-18 rejection works
- ☐ User Data Rights Features
  - ☐ Data export endpoint (GET /users/me/data-export)
    - ☐ Download user data in machine-readable format
    - ☐ Includes all personal data
    - ☐ Available within 30 days of request
  - ☐ Account deletion endpoint (DELETE /users/me/account)
    - ☐ Anonymize user data
    - ☐ Delete personal identifiers
    - ☐ Retain minimal data for legal compliance
    - ☐ Available immediately
  - ☐ Data rectification interface
    - ☐ Users can edit their data
    - ☐ Changes reflected immediately
- ☐ Consent & Tracking
  - ☐ Cookie consent banner (before Analytics)
  - ☐ Consent stored with timestamp
  - ☐ User consent acceptance at registration
  - ☐ Privacy acceptance checkbox required
  - ☐ TOS acceptance checkbox required

- ☐ Marketing consent checkbox (optional)
- ☐ All consent decisions logged with IP & timestamp
- ☐ Data Processing
  - ☐ Create Data Processing Register
    - ☐ Document all data collected
    - ☐ Document processing purposes
    - ☐ Document retention periods
    - ☐ Document third parties accessing data
  - ☐ Data Processing Agreements with vendors
    - ☐ Cloudinary (image storage)
    - ☐ Resend (email service)
    - ☐ Sentry (error monitoring)
    - ☐ All third parties handling data
  - ☐ Privacy by Design
    - ☐ Collect minimum necessary data
    - ☐ Anonymize when possible
    - ☐ Secure storage and transmission

#### **Before Launch - Documentation:**

- ☐ Data Retention Policy defined
  - ☐ How long data retained
  - ☐ When/how data deleted
  - ☐ Deletion process documented
- ☐ Breach Notification Procedure
  - ☐ Incident response plan
  - ☐ ANSPDCP notification process
  - ☐ User notification procedures
  - ☐ Timeline for notifications
- ☐ GDPR Compliance Checklist
  - ☐ Lawful basis identified (legitimate interest)
  - ☐ Privacy Notice provided
  - ☐ Consent mechanisms working
  - ☐ User rights procedures documented
  - ☐ Third-party clauses in place

#### **Data Protection Authority Registration**

##### **Within 3 Months of Launch:**

- ☐ ANSPDCP registration (if processing large scale)
  - ☐ Determine if registration required
  - ☐ Complete ANSPDCP registration form
  - ☐ Pay registration fee if applicable
  - ☐ Confirmation received



## Intellectual Property

### Before Launch (Optional but recommended):

- ☐ Trademark search at OSIM (Romanian IP office)
  - ☐ "SwapBuds" - search for existing trademarks
  - ☐ Similar names check
  - ☐ Document findings

### After Funding (Year 2+):

- ☐ OSIM trademark application (€500-2000)
- ☐ EU trademark through EUIPO (€850, broader protection)
- ☐ Domain registrations (.ro, .com, .eu)

## Insurance

### Before Public Launch:

- ☐ Professional Liability Insurance (RCA Profesional)
  - ☐ €500-1000/year
  - ☐ Covers professional mistakes
- ☐ Cyber Liability Insurance (Asigurare Cyber)
  - ☐ €1000-2000/year
  - ☐ Covers data breaches, hacking
  - ☐ Critical for marketplace with user data
- ☐ Get quotes from:
  - ☐ Allianz-Țiriac
  - ☐ Generali România
  - ☐ Groupama
  - ☐ Asiom

## Legal Review & Consultation

### Before Public Launch:

- ☐ Have lawyer review all documents
  - ☐ Cost: €3000-5000 for template review
  - ☐ Find lawyer: Baroul Bucuresti, [TechLaw.ro](http://TechLaw.ro)
  - ☐ Timeline: 2-3 weeks for review

### As Needed:

- ☐ Consult lawyer on legal issues
  - ☐ Standard rate: €150-600/hour
  - ☐ Have clear questions prepared
  - ☐ Build ongoing relationship with lawyer

## PART 5: PRE-LAUNCH CHECKLIST (Final Verification)

### ✓ Code Quality & Testing (Week 10-11)

#### Unit & Integration Tests

- ☐ 484 backend tests passing
- ☐ Frontend E2E tests passing
- ☐ 80%+ code coverage (both backend and frontend)
- ☐ Zero failing tests before deployment
- ☐ Test failures documented and addressed

#### Code Quality

- ☐ TypeScript strict mode enabled, no errors
- ☐ ESLint: Zero errors, zero warnings
- ☐ Prettier formatting consistent across codebase
- ☐ No console.logs in production code
- ☐ No debug/test code remaining
- ☐ Unused imports/exports removed
- ☐ Dead code removed

#### Dependency Management

- ☐ npm/yarn audit: ZERO vulnerabilities
- ☐ No outdated major versions
- ☐ Dependencies updated to stable versions
- ☐ Changelog reviewed for breaking changes
- ☐ Peer dependencies resolved correctly

### 🔒 Security Hardening (Week 10)

#### Authentication & Authorization

- ☐ Password hashing with bcrypt verified
- ☐ JWT token generation secure (32+ char secret)
- ☐ JWT refresh token rotation implemented
- ☐ Session timeout configured (15-30 min idle)
- ☐ Logout clears all tokens
- ☐ Rate limiting on login attempts (5 attempts, 15 min lockout)
- ☐ All authentication events logged

#### API Security

- ☐ CORS configuration locked down (no wildcard \*)
- ☐ Only specific origins allowed
- ☐ Credentials sent with requests properly handled
- ☐ Rate limiting on all public endpoints
  - ☐ General: 100 requests/min per IP
  - ☐ Auth: 10 requests/min per IP

- ☐ Search: 50 requests/min per IP
- ☐ API keys rotated every 90 days

### **Input Validation & Sanitization**

- ☐ All user inputs validated server-side
- ☐ No client-side validation as sole protection
- ☐ SQL injection prevention verified (Prisma handles this)
- ☐ XSS protection enabled (sanitize HTML inputs)
- ☐ CSRF token validation on state-changing endpoints
- ☐ File upload validation:
  - ☐ File type check (image/\* only)
  - ☐ File size limits enforced (< 10MB)
  - ☐ Virus scanning if possible
  - ☐ Rename files to prevent execution

### **Data Protection**

- ☐ Sensitive data encrypted at rest (passwords, tokens)
- ☐ SSL/TLS for all data in transit
- ☐ HTTPS enforced (no HTTP allowed)
- ☐ Environment variables never logged
- ☐ API responses don't leak sensitive info (IDs, internal errors)
- ☐ Error messages generic (no stack traces to users)
- ☐ Secrets not committed to git

### **Headers & Security Policies**

- ☐ Security headers implemented (Helmet.js)
  - ☐ Content-Security-Policy (CSP)
  - ☐ X-Content-Type-Options
  - ☐ X-Frame-Options
  - ☐ X-XSS-Protection
  - ☐ Strict-Transport-Security (HSTS)
- ☐ CORS headers properly set
- ☐ Referrer-Policy configured
- ☐ Permissions-Policy configured

### **Audit Logging**

- ☐ All authentication attempts logged
- ☐ All sensitive actions logged (deletes, exports, etc.)
- ☐ Logs include timestamp, user ID, IP address, action
- ☐ Logs not exposed via API
- ☐ Audit trail accessible for compliance

## ▮ GDPR & Legal Compliance (Week 10)

### Age Verification

- ☐ Date of birth field required at registration
- ☐ Age automatically calculated
- ☐ Users under 18 rejected at signup
- ☐ Error message: "You must be 18 or older to use SwapBuds"
- ☐ Testing: Create test account with birth date 2010, verify rejection

### Consent & Acceptance

- ☐ Cookie consent banner displays on first visit
- ☐ Users can accept/reject all cookies
- ☐ Consent decision stored in localStorage
- ☐ Stored with timestamp, user consent tracked in DB
- ☐ Privacy Policy link in banner
- ☐ Cookie Policy link in banner
- ☐ ToS acceptance required at registration checkbox
- ☐ Privacy Policy acceptance required at registration checkbox
- ☐ Marketing consent checkbox (optional)
- ☐ All consents logged with user ID, IP, timestamp

### User Data Rights

- ☐ Data export endpoint (GET /users/me/data-export)
  - ☐ Returns all personal data as JSON/CSV
  - ☐ Includes: profile, items, trades, messages, reviews
  - ☐ Includes: consent records, login history
  - ☐ Within 30 days GDPR requirement
  - ☐ Testing: Export own data, verify completeness
- ☐ Account deletion endpoint (DELETE /users/me/account)
  - ☐ Soft delete: Mark user as deleted
  - ☐ Anonymize personal data: Remove name, email, phone
  - ☐ Retain trade history (anonymized) for dispute resolution
  - ☐ Remove authentication tokens
  - ☐ Delete session data
  - ☐ Testing: Delete account, verify data anonymized
- ☐ Data rectification
  - ☐ Users can edit profile, email, phone
  - ☐ Changes immediately reflected
  - ☐ History of changes not exposed (privacy)

## Data Retention Policy

- ☐ Active users: Data retained while account active + 30 days after deletion
- ☐ Deleted users: Anonymized data retained 1 year for legal compliance
- ☐ Trade disputes: Data retained for dispute resolution period + 1 year
- ☐ Backups: Retained 30 days as per backup retention
- ☐ Access logs: Retained 90 days for security
- ☐ Email logs: Retained 90 days for troubleshooting

## Legal Documents

- ☐ Terms of Service:
  - ☐ Published on website, linked in footer
  - ☐ Available in Romanian language
  - ☐ Covers: User responsibilities, dispute resolution, content moderation
  - ☐ Lawyer reviewed (Romanian law specialist)
- ☐ Privacy Policy:
  - ☐ Published on website, linked in footer
  - ☐ Romanian language primary
  - ☐ Covers: Data collection, processing, storage, rights
  - ☐ GDPR compliant (Articles 13, 14)
  - ☐ Lawyer reviewed
- ☐ Cookie Policy:
  - ☐ Published on website, linked in banner
  - ☐ Lists all cookies/trackers
  - ☐ Explains purpose of each
  - ☐ User consent mechanism working
- ☐ Community Guidelines:
  - ☐ Available in app and website
  - ☐ Defines prohibited conduct
  - ☐ User reporting mechanism
  - ☐ Moderation process

## GDPR Compliance Audit

- ☐ Lawful basis for data processing identified: ✓ Legitimate interest (community building)
- ☐ Privacy Notice provided: ✓ Privacy Policy
- ☐ Consent mechanisms working: ✓ All tested
- ☐ User rights accessible: ✓ Export, delete, rectify
- ☐ Data Protection Agreement with processors: ✓ Cloudinary, Resend, Sentry
- ☐ Data Processing Register maintained: ✓ Current and updated
- ☐ Breach notification procedure documented: ✓ Within 72 hours to ANSPDCP

## 🔧 Infrastructure & Deployment (Week 10-11)

### Database

- ☐ PostgreSQL connection string configured
- ☐ Database accessible from production server
- ☐ Connection pooling configured (10-20 connections)
- ☐ SSL/TLS encryption enabled
- ☐ All migrations run successfully
- ☐ Database backup created and tested
- ☐ Restoration procedure documented

### Redis Cache

- ☐ Redis connection configured
- ☐ Connection pooling working
- ☐ Cache keys are namespaced
- ☐ Expiration times set appropriately
- ☐ Memory limits configured
- ☐ Backup strategy for cache (if needed)

### File Storage

- ☐ Cloudinary account connected
- ☐ Image upload works end-to-end
- ☐ Image transformations working (resize, optimize)
- ☐ CDN serving images globally
- ☐ Backup mechanism understood
- ☐ Unused images cleanup scheduled

### Email Service

- ☐ Resend or AWS SES configured
- ☐ SMTP credentials secure in environment variables
- ☐ Test email sends successfully
- ☐ From address configured correctly
- ☐ Reply-to address set
- ☐ Email templates tested (registration, password reset, trade updates)
- ☐ Email deliverability verified
- ☐ Bounce handling configured

### Monitoring & Alerting

- ☐ Sentry connected to backend
- ☐ Sentry connected to frontend
- ☐ Test error sent and received
- ☐ BetterStack uptime monitoring active
- ☐ Alert channels configured (email at minimum)
- ☐ Test alert sent successfully
- ☐ Dashboard accessible and understood

## Environment Variables

- ☐ All secrets in environment variables (not git)
- ☐ Production secrets different from development
- ☐ .env.example file created for reference (no real values)
- ☐ Environment variable list documented
- ☐ Secrets rotation schedule established
- ☐ GitHub secret scanning enabled

## CI/CD Pipelines

- ☐ Backend CI workflow tested
  - ☐ Tests run on PR
  - ☐ Linting passes
  - ☐ TypeScript builds
  - ☐ Coverage reports generated
- ☐ Frontend CI workflow tested
  - ☐ Linting passes
  - ☐ Type checking passes
  - ☐ Build succeeds
  - ☐ E2E tests pass
- ☐ Deployment workflow tested
  - ☐ Merge to main triggers deploy
  - ☐ Backend deploys successfully
  - ☐ Frontend deploys successfully
  - ☐ No downtime or partial failures

## SSL/TLS Certificates

- ☐ HTTPS enabled on all domains
- ☐ SSL certificate valid (not expired)
- ☐ Certificate covers all subdomains
- ☐ Certificate auto-renewal configured (Let's Encrypt or similar)
- ☐ HTTP redirects to HTTPS
- ☐ HSTS header enabled

## Domain & DNS

- ☐ Domain registered ([swapbuds.ro](https://swapbuds.ro))
- ☐ DNS records configured correctly
  - ☐ A record points to correct IP
  - ☐ CNAME records for subdomains
  - ☐ MX records for email
  - ☐ TXT records for SPF, DKIM (if applicable)
- ☐ DNS propagation verified
- ☐ Domain email working (if using for notifications)

## ▮ Performance & Optimization (Week 10-11)

### Backend Performance

- ☐ API response time: < 200ms average
- ☐ Database queries: < 100ms
- ☐ Slow query log reviewed and optimized
- ☐ N+1 queries eliminated
- ☐ Database indexes on common queries
- ☐ Pagination implemented on list endpoints
- ☐ Caching strategy implemented for frequent queries
- ☐ Load testing completed (1000+ concurrent users)

### Frontend Performance

- ☐ Page load time: < 3 seconds
- ☐ Core Web Vitals acceptable
  - ☐ Largest Contentful Paint (LCP): < 2.5s
  - ☐ First Input Delay (FID): < 100ms
  - ☐ Cumulative Layout Shift (CLS): < 0.1
- ☐ Lighthouse score: 90+
- ☐ Bundle size optimized (< 500KB gzipped)
- ☐ Code splitting implemented
- ☐ Images optimized and lazy-loaded
- ☐ Service worker working for offline support

### Database Performance

- ☐ Query execution plans reviewed
- ☐ Indexes created on foreign keys
- ☐ Indexes created on frequently queried fields
- ☐ Connection pool size appropriate
- ☐ Query caching working
- ☐ Database size monitored

## ▮ Documentation (Week 11)

### API Documentation

- ☐ Swagger/OpenAPI spec complete
- ☐ All endpoints documented
- ☐ Request/response examples included
- ☐ Authentication documented
- ☐ Error codes documented
- ☐ Rate limits documented
- ☐ Swagger UI accessible at /api/docs



## Code Documentation

- ☐ README.md in both repos
  - ☐ Setup instructions
  - ☐ Development environment setup
  - ☐ Testing instructions
  - ☐ Deployment instructions
- ☐ Complex functions have comments
- ☐ Inline comments explain "why" not "what"

## Deployment Documentation

- ☐ Deployment process documented
- ☐ Pre-deployment checklist
- ☐ Deployment procedure step-by-step
- ☐ Post-deployment verification steps
- ☐ Rollback procedure documented
- ☐ Emergency contact list

## Troubleshooting Guide

- ☐ Common issues documented
- ☐ Resolution steps for each issue
- ☐ Support contact information
- ☐ Escalation procedures

## User Documentation

- ☐ Help center / FAQ published
- ☐ How-to guides for main features
- ☐ Troubleshooting guide for users
- ☐ Contact support procedure
- ☐ FAQ covers: account creation, trading, disputes, safety

## Changelog / Release Notes

- ☐ Version history documented
- ☐ Features added listed
- ☐ Bugs fixed listed
- ☐ Known issues documented
- ☐ Upgrade path clear for users

## ▮ Marketing Readiness (Week 11-12)

### Website & Content

- ☐ Homepage optimized for conversion
- ☐ Meta tags (title, description) set
- ☐ Open Graph images configured
- ☐ Twitter Card configured
- ☐ Landing page CTA clear

- ☐ Landing page loads quickly
- ☐ Mobile responsive
- ☐ All links working

## **Social Media**

- ☐ All accounts created and branded
- ☐ Profile pictures consistent
- ☐ Bios complete with CTA
- ☐ Verified badges if possible
- ☐ First week of content scheduled (20+ posts)
- ☐ Content calendar created for next month
- ☐ Hashtag strategy documented

## **Email Marketing**

- ☐ Email service account created
- ☐ List of early signups exported
- ☐ Welcome sequence created (5 emails)
- ☐ Launch announcement email ready
- ☐ Email design templates created
- ☐ Unsubscribe mechanism working

## **Analytics**

- ☐ Google Analytics configured
- ☐ Tracking code installed
- ☐ Events tracked (signup, first trade, etc.)
- ☐ Goals configured
- ☐ Dashboard created
- ☐ Baseline metrics recorded

## **Press & Media**

- ☐ Press release written and ready
- ☐ Press kit created
- ☐ Company bio written
- ☐ Founder bio written
- ☐ Press list of 50+ contacts
- ☐ Pitch emails drafted

## **Product Hunt**

- ☐ Account created and verified
- ☐ Product page set up (drafts)
- ☐ Product description written (100-200 words)
- ☐ Images/screenshots uploaded
- ☐ Video thumbnail prepared
- ☐ Tagline finalized
- ☐ Supporters lined up (friends, advisors)

## ▯ Launch Readiness (Day Before Launch)

### Final Technical Check

- ☐ All systems online and responding
- ☐ Database backups fresh
- ☐ Error monitoring working
- ☐ Logs accessible
- ☐ All emails sending
- ☐ File uploads working
- ☐ Real-time features working (WebSockets/messaging)
- ☐ APIs responding within SLA
- ☐ Frontend loads and functions correctly
- ☐ Mobile app responsive

### Final Marketing Check

- ☐ Social media posts scheduled and queued
- ☐ Email launch sequence ready
- ☐ Product Hunt set to go live
- ☐ Press contacts ready
- ☐ Team available for launch day support
- ☐ Communication channels open (Discord, Twitter, email)
- ☐ Response plan for issues documented

### Final Business Check

- ☐ Legal documents published on website
- ☐ Terms of Service displayed
- ☐ Privacy Policy displayed
- ☐ Cookie consent working
- ☐ Support email configured
- ☐ Support contact person assigned
- ☐ Escalation procedures ready

### Launch Day Preparation

- ☐ Team available (or just you!) for monitoring
- ☐ Communication channels monitored:
  - ☐ Twitter/X mentions
  - ☐ Email support
  - ☐ Discord channel
  - ☐ Direct messages
  - ☐ Product Hunt comments
- ☐ Error monitoring active and alerts configured
- ☐ Status page ready for updates
- ☐ Rollback plan accessible
- ☐ Emergency contact list prepared

PART 6: CRITICAL SUCCESS FACTORS FOR SOLO DEVELOPER

▮ Key Principles

- 1. **Ship, Don't Perfect** - Done is better than perfect. Iterate based on user feedback.
- 2. **MVP First** - Only core features for launch. Nice-to-haves can come later.
- 3. **Test Early** - Friends/family beta before public launch.
- 4. **Iterate Fast** - Fix issues immediately. Users are your QA team.
- 5. **Stay Lean** - Free tier infrastructure as long as possible.
- 6. **Focus** - Romanian market only initially. Master one market before expanding.
- 7. **Listen** - User feedback drives roadmap.
- 8. **Document** - Write down what works and what doesn't.
- 9. **Celebrate** - Small wins matter. Pat yourself on the back.
- 10. **Sustainable Pace** - Marathon, not sprint. You've got this!

▮ Success Metrics (By Milestone)

| Milestone | Users  | Trades | Retention   | Satisfaction     |
|-----------|--------|--------|-------------|------------------|
| Week 4    | 0-10   | 0      | N/A         | Feedback quality |
| Week 8    | 10-50  | 5-10   | Good        | Beta feedback    |
| Week 12   | 50-100 | 50+    | Day 30: 40% | NPS 40+          |
| Week 16   | 500    | 500+   | Day 30: 50% | NPS 50+          |
| Month 4   | 1,000  | 1,000+ | Day 30: 50% | NPS 55+          |

⚠ Common Pitfalls to Avoid

- 1. **Over-engineering** - Don't build for scale before launch. YAGNI.
- 2. **Feature creep** - Stick to MVP. Say "no" to nice-to-haves.
- 3. **Neglecting legal** - Don't skip compliance. It's not optional.
- 4. **Poor testing** - Test everything. Especially edge cases.
- 5. **No monitoring** - Deploy monitoring before issues arise.
- 6. **Ignoring feedback** - Users know best. Listen and iterate.
- 7. **Burning out** - Take breaks. Health first, product second.
- 8. **No backup plan** - Always have a disaster recovery plan.
- 9. **Skipping documentation** - Future you will thank present you.
- 10. **Giving up too early** - Growth takes time. Compound effort.

CONCLUSION

This comprehensive plan provides a complete roadmap for launching SwapBuds as a solo developer. The 16-week timeline is ambitious but achievable. Key to success:

- 1. **Execute Phase 1 (Weeks 1-6)** - Build functional MVP with legal compliance
- 2. **Test Phase 2 (Weeks 7-10)** - Deploy to free tier, gather feedback
- 3. **Iterate Phase 3 (Weeks 11-14)** - Beta test with real users, refine product
- 4. **Launch Phase 4 (Weeks 15-16)** - Go public with coordinated marketing push

**Remember:** You built the entire backend (v1.0, 484 tests) in 3 days. You can absolutely build this frontend and launch successfully. The hardest part is already done. You've got this! 📦

**Last Updated:** November 29, 2025

**Next Review:** Weekly during launch preparation, then daily during launch week